

Strategic Technical Architecture for a Next-Generation Corporate Aviation Maintenance Management System

1. Executive Overview and Operational Scope

The aviation industry stands at a critical juncture where the digitization of flight operations and maintenance is no longer a luxury but a regulatory and operational necessity. Corporate aviation operators, specifically those governed by **14 CFR Part 135** (Charter) and **Part 91** (Private Operations), face a unique set of challenges: high-tempo operations, diverse fleet types, and the imperative to maintain "airline-level" safety standards with leaner teams. The objective of this technical report is to outline the comprehensive architecture for a proprietary web application designed to serve as a **Continuing Airworthiness Management System**, herein referred to as "**AviaSync**".

AviaSync is engineered to rival industry incumbents like **TrustFlight** by offering a seamless, closed-loop workflow that integrates Pilot Defect Reporting (Squawks), Maintenance Execution, and Regulatory Compliance into a single "Source of Truth." A central differentiator of this architecture is its deep integration with **CAMP Systems**—the ubiquitous maintenance tracking provider—and the implementation of a proprietary **Predictive Analytics Engine**. This engine will not merely log historical data but will utilize machine learning algorithms to forecast maintenance task durations and flag component failure intervals before they result in Aircraft on Ground (AOG) events.

The architecture proposed herein adheres strictly to **FAA Advisory Circular 120-78B** regarding electronic signatures and recordkeeping ¹, ensuring that the digital workflow remains legally defensible. It enables a paperless transition where pilots can report issues, maintenance teams can defer items via a digital **Minimum Equipment List (MEL)**, and airworthiness releases are synchronized instantly across the fleet. Furthermore, the system addresses the critical human factors of maintenance through a rigorous **Shift Handover Protocol**, designed to mitigate the communication errors that account for a significant percentage of aviation incidents.³

2. Regulatory Compliance Framework and Digital Certification

In the domain of aviation software, code quality is synonymous with regulatory compliance. A web app for corporate aviation is not simply a data entry tool; it is a legal instrument used to determine the airworthiness of an aircraft. Therefore, the foundational layer of this application must be built upon the specific requirements set forth by the FAA and EASA.

2.1 Electronic Recordkeeping and Signatures (FAA AC 120-78B)

To legally replace the physical paper logbook and maintenance forms, the system must meet the "Acceptance" criteria of **AC 120-78B**.⁴ This advisory circular dictates the technical standards for electronic signatures and recordkeeping systems.

2.1.1 Cryptographic Signature Architecture

The system must distinguish between a user "logging in" and a user "signing" a record. A simple session cookie is insufficient for an airworthiness release. The architecture requires a **Step-Up Authentication** mechanism.

- **Intent to Sign:** The User Interface (UI) must clearly display the text of the record being signed (e.g., "I certify that the work described above has been performed in accordance with...") and require an affirmative action distinct from data entry.⁵
- **Credentialing:** The signature event will require the re-entry of a password or a biometric verification (via WebAuthn) to generate a unique cryptographic hash.
- **Non-Repudiation:** The system will generate a **SHA-256 hash** of the record's content (the squawk description, the corrective action, the date, and the user ID). This hash is stored alongside the record. If the record is altered in the database by even a single bit, the hash will fail validation, alerting the system to a data integrity breach. This satisfies the requirement that the signature cannot be excised or copied to another record.⁴

2.1.2 Audit Trails and Immutability

FAA regulations require that once a maintenance action is recorded and signed, it must be permanent. The database design will utilize an **Append-Only Ledger** approach for the MaintenanceLog and FlightLog tables.

- **Version Control:** If a technician makes an error (e.g., enters the wrong part number), the system does not UPDATE the row. Instead, it creates a new row referencing the original as superseded.
- **Traceability:** The Audit_Log table will capture every interaction: User_ID, Timestamp_UTC, IP_Address, Action_Type, Pre_Value, and Post_Value. This allows an FAA inspector to replay the history of any specific maintenance event.⁷

2.2 EASA Part-M and International Interoperability

For operators flying into Europe or managing aircraft under EASA jurisdiction, the system must align with **Part-M** (Continuing Airworthiness) and **Part-ML** (Light Aircraft) standards.⁸

- **Accessibility:** EASA mandates that records be "readily available" to the competent authority. The cloud infrastructure will be deployed across multiple availability zones (AWS Multi-AZ) to ensure 99.99% uptime.
- **Pilot-Owner Maintenance:** The Role-Based Access Control (RBAC) system must accommodate "Pilot-Owner" permissions, allowing pilots to sign off on specific, limited maintenance tasks (e.g., changing a bulb or updating navigation databases) as defined in **Appendix II to Part-ML**, distinct from the permissions of a certified A&P mechanic.¹⁰

2.3 Managing the Station Copy and "Print-Off" Capability

While the goal is a fully digital workflow, regulations regarding the "**Station Copy**" (the record left on the ground prior to departure) must be addressed.

- **Digital Synchronization:** In a fully connected environment, the cloud database serves as the Station Copy. Data is synced immediately upon pilot signature.
- **Offline Contingency:** The requirement to "print it off" is critical for remote operations or ramp checks by inspectors who demand physical proof. The system will feature a **PDF Generation Engine** that renders the digital log data into a format identical to the operator's legacy paper forms. This PDF can be sent to a portable Bluetooth thermal printer or emailed directly to the local FBO (Fixed Base Operator) for printing and retention, ensuring compliance with **14 CFR 91.417**.¹¹

3. The Electronic Technical Log (ETL) Pilot Workflow

The Electronic Technical Log (ETL) is the primary data ingestion point for the system. It replaces the carbon-copy logbook in the cockpit. The design philosophy here is "Guided Compliance"—using software constraints to prevent pilot error and ensure that every squawk entered is actionable and standardized.

3.1 Structured Squawk Entry and ATA Coding

One of the greatest inefficiencies in maintenance is deciphering vague pilot reports (e.g., "Something creates a noise"). To solve this, the Pilot App will utilize a **Hierarchical Fault Tree** based on **ATA iSpec 2200** codes.¹³

The Workflow:

1. **Fault Identification:** The pilot selects "Report Defect." The system presents a visual model of the aircraft or a system list.
 - *Pilot Selects:* "24 - Electrical Power" -> "24-30 - DC Generation" -> "Generator 1".
2. **Smart Form Logic:** Based on the ATA code selection, the form dynamically adjusts. If "Tires" are selected, the form requests "Tread Depth" and "Position." If "Engine" is selected, it requests "N1/N2 values" and "Oil Pressure." This ensures maintenance receives the specific data needed for troubleshooting without back-and-forth communication.
3. **Phase of Flight:** The system automatically captures the aircraft state (On Ground, Taxi, Climb, Cruise) from the avionics bus (if integrated via ARINC gateway) or requires pilot selection. This data is vital for the predictive analytics module, as vibration-related failures often correlate with specific flight phases.¹⁴

3.2 Digital MEL Integration and Deferral Logic

The prompt specifically requests that "maintenance team gets this... process it... send back the MEL digitally." This is the core of the **Deferral Workflow**.

3.2.1 The Master Minimum Equipment List (MMEL) Database

The backend will host a digitized version of the FAA/EASA Approved **Master Minimum Equipment List (MMEL)**. This database maps ATA codes to deferral categories and procedures.¹³

3.2.2 The Deferral Interaction

When a pilot logs a squawk (e.g., "Right Landing Light Inop"), the system performs a real-time query against the MEL database.

1. **Automated Assessment:** The system checks: *Is this item in the MEL?*
 - **Yes:** The system displays the **(O) Operational** and **(M) Maintenance** procedures associated with the item.
 - **No:** The system flags the item as "No-Go" (AOG), preventing dispatch until rectified.
2. **Maintenance Action (The "Process It" Step):**
 - The maintenance controller receives the squawk instantly on the Manager Dashboard.
 - They review the MEL reference. If **(M) Procedures** (e.g., "Deactivate and Secure breaker") are required, the system generates a **Provisional Deferral Work Order**.
 - A mechanic performs the (M) procedure and digitally signs the work order.
3. **Digital Return to Cockpit:**
 - Once the mechanic signs the (M) procedure, the system updates the aircraft status.
 - The pilot's device receives a push notification: **"MEL Applied: Right Landing Light - Category C (10 Days). Expires:."**
 - The digital "Placard" is visualized on the pilot's dashboard, overlaying the aircraft diagram.¹⁶

3.3 The Airworthiness Release (Workflow Closure)

The final step in the pilot workflow is the acceptance of the aircraft. The app will present a summary screen:

- **Open Squawks:** None.
- **Deferred Items:** 1 (Right Landing Light).
- **Scheduled Maintenance:** None Due.
- **Action:** The pilot signs the "Acceptance of Aircraft," acknowledging the deferred items. This signature closes the loop and generates the official flight log record.⁸

4. Maintenance Management and CAMP Systems Integration

The user specifically requested the ability to "pull job cards from CAMP" and "track data." While the ETL handles the *unplanned* work (squawks), the Maintenance Management module handles the *planned* work (inspections). This system acts as a high-fidelity interface for the CAMP database, enhancing it with granular workflow management.

4.1 CAMP API Integration Strategy

Integration with **CAMP Systems** (and potentially others like CORRIDOR or CMP) requires a robust middleware layer capable of translating external API schemas into our internal data model.

4.1.1 API Architecture and Endpoints

We will utilize the **CAMP Connect / MTX API** to establish a bidirectional sync.¹⁸

- **Authentication:** The system will manage OAuth2 tokens or API keys provided by CAMP, securely stored in an AWS Secrets Manager.
- **Data Ingestion (Pull):** A scheduled task (running every 1-4 hours) will call the GetDueList and GetTaskCard endpoints for every aircraft in the fleet.
 - **Endpoint:** GET /v1/aircraft/{tailNumber}/due-items
 - **Payload:** Returns a JSON list of tasks approaching their interval (e.g., "12 Month Inspection," "500 Hour Lubes").
 - **Job Card Content:** The system will fetch the specific procedural text and PDF attachments for each task via GET /v1/task/{taskId}/details.²⁰
- **Compliance Sync (Push):** When a technician signs off a task in our app, the system queues a POST request to CAMP to update the official record.
 - **Endpoint:** POST /v1/compliance
 - **Payload:** { "TaskId": "12345", "Date": "2026-02-14", "Hours": 1450.5, "Cycles": 890, "Technician": "ID_55" }.

4.1.2 Handling "Manual" Non-CAMP Tasks

To create a complete history, the system must handle tasks that CAMP does not track (e.g., "Paint touch-up," "Install new coffee maker").

- **Unified Schema:** The database will use a polymorphic relationship for Tasks. A Task can be of type CAMP_Source (linked to an external ID) or Internal_Source (generated locally).
- **Historical Retention:** All internal tasks are stored in the MaintenanceHistory table. This allows the operator to search for "Coffee Maker" and see every time it was repaired, creating a complete "cradle-to-grave" service history that supplements the official regulatory log.²¹

4.2 Advanced Work Order Management

The user requires a display to "see what work orders are in progress and be able to assign priority." This is the **Maintenance Control Center (MCC)** dashboard.

4.2.1 The Kanban Work Order Board

The UI will utilize a **Kanban** visualization (columns: *Pending*, *Assigned*, *In Work*, *Quality Control*, *Closed*) to represent the lifecycle of a Work Order (WO).

- **Grouping:** The Manager can select multiple CAMP tasks (e.g., Task A, Task B, Task C)

and group them into a single WO package (e.g., "WO-2026-005: 100 Hour Check").

- **Priority Triage:** The Manager can tag WOs as "Routine," "Urgent," or "AOG."
 - **Logic:** Marking a WO as "AOG" triggers a push notification to all assigned technicians and moves the card to the top of the mobile list with a distinct red visual indicator.²¹

4.2.2 Team and Individual Assignment

The database will model WorkGroups (e.g., "AM Shift," "PM Shift," "Avionics Team") and Users (Individual Techs).

- **Assignment Logic:** The Manager can drag a WO onto the "AM Shift" group. The system logic then propagates this assignment to all users tagged with Shift: AM.
- **Granularity:** For critical inspections (RII - Required Inspection Items), the manager can assign a specific individual inspector to the "QC" step of the card, ensuring regulatory separation of duties (the person doing the work cannot sign the inspection).²²

5. Technician Execution and Shift Handoff Protocol

The technician's interface is designed for mobility and execution. It runs on ruggedized tablets (iPad/Android) and handles the "start and end job times" and "hand off" requirements.

5.1 Real-Time Job Tracking

Accurate time tracking is the fuel for the predictive analytics module.

- **Execution Flow:**
 1. Technician logs in via biometric auth.
 2. Selects "My Jobs."
 3. Taps **"Clock On"** for a specific task. The system records start_timestamp.
 4. Technician performs work, viewing the CAMP job card PDF directly in the app.
 5. Taps **"Clock Off"** or **"Pause."** If pausing, a reason must be selected (e.g., "Waiting for Parts," "Shift End"). This distinguishes between *elapsed time* and *touch time*.²¹

5.2 The "5/40" Shift Handover Protocol

Research indicates that 40% of maintenance accidents occur during the 5% of time spent on shift handovers (the "5/40" problem).³ To address this, the system implements a **Structured Digital Handover**.

Technical Implementation:

1. **Blocker:** The system prevents a technician from clocking out of a shift if they have "In Progress" tasks without a handover note.
2. **Handover Record:** The technician must enter a status update for every open task (e.g., "Cowling removed, filter housing drained, waiting on O-ring").
 - **Photos:** The tech can attach a photo of the current state.
3. **Incoming Acknowledgement:** When the PM Shift technician clocks into that task, they are presented with the "Handover Record." They must digitally sign "Acknowledged" before they can start their timer. This establishes a clear chain of custody and ensures

the incoming tech is fully aware of the aircraft's state.²³

6. Predictive Analytics and Data Intelligence

The requirement to "predict how long tasks will take" and "flag intervals when things typically become an issue" moves this system from a passive logger to an active intelligence tool.

6.1 Predictive Task Duration Modeling

We will implement a **Supervised Machine Learning Model** (specifically, a **Random Forest Regressor**) to predict task duration. Random Forest is chosen for its ability to handle non-linear relationships and resistance to overfitting with relatively small datasets (common in small fleets).²⁵

Feature Engineering (Input Variables):

To train the model, we will extract the following features from the database:

1. **Task Type:** (e.g., "Wheel Change," "Borescope Inspection").
2. **Aircraft Model & Age:** (e.g., "G550, 10,000 hours"). Older aircraft often require longer repair times due to corrosion or seized fasteners.
3. **Technician Skill Level:** A "Level 3" tech is faster than a "Level 1."
4. **Historical Average:** The rolling mean of "Touch Time" for this task over the last 100 executions.
5. **Concurrent Work:** Is this task part of a heavy check (crowded hangar) or a line maintenance event?

Algorithm Execution:

$$\text{Estimated Duration} = f(\text{Task}, \text{Age}, \text{Skill}, \text{History})$$

When a Manager adds a task to a Work Order, the model runs this inference. The UI displays two numbers:

- **Standard Book Time:** (From CAMP/OEM Manual, e.g., 4.0 Hours).
- **Predicted Actual Time:** (From ML Model, e.g., 5.2 Hours).
- **Insight:** If the difference is >20%, the system flags a "Schedule Risk," allowing the manager to adjust the aircraft's promised delivery time.²⁶

6.2 Interval Flagging and Predictive Maintenance

To "look for intervals when things typically become an issue," the system will utilize **Weibull Analysis**, a statistical method used in reliability engineering to determine the failure distribution of components.²⁷

Technical Mechanism:

1. **Data Harvesting:** The system aggregates "Unscheduled Squawks" (failures) for specific components (e.g., "Starter Generator") across the fleet history.
2. **Survival Function:** The algorithm calculates the **Hazard Rate** $\lambda(t)$ for the component type. It identifies the "Infant Mortality" phase, "Useful Life" phase, and "Wear-Out" phase.
3. **The "Early Flag" (Proactive Alerting):**

- *Scenario:* Historical data shows that "Fuel Pump A" on this aircraft type has a rising failure rate starting at 1,200 Flight Hours (FH).
- *Monitoring:* The system tracks the current FH of all Fuel Pumps in the fleet.
- *Trigger:* When Aircraft N101's pump hits 1,150 FH, the system generates a **"Predictive Alert"** on the Manager Dashboard: *"Component Risk Warning: Fuel Pump A approaching high-failure interval. Recommendation: Schedule replacement during upcoming maintenance window."*
- This capability allows the manager to order the part and schedule the work *before* the component fails and causes an AOG event.²⁸

7. System Architecture and Technology Stack

To support these real-time, data-intensive features, the underlying technology stack must be robust, scalable, and secure.

7.1 Tech Stack Selection

Component	Technology	Justification
Frontend (Web)	React.js + TypeScript	Component-based structure ideal for complex dashboards and interactive Gantt charts. Type safety reduces runtime errors. ³¹
Frontend (Mobile)	React Native	Allows for a single codebase to deploy native apps to iOS/Android tablets with access to local storage for offline MEL databases. ³²
Backend API	Node.js (NestJS)	Scalable, modular architecture. Excellent support for handling asynchronous I/O (vital for real-time WebSocket connections).
Database (Relational)	PostgreSQL	ACID compliance is non-negotiable for regulatory records. Stores User, Aircraft, and Work Order tables.
Database (Time-Series)	TimescaleDB	Optimized for storing high-volume sensor data and "Time Tracking" logs needed for the predictive engine. ²⁸
Real-Time Layer	Socket.IO (WebSockets)	Enables instant

		synchronization. When a pilot signs a log, it appears on the Manager's screen immediately without refreshing. ³³
Infrastructure	AWS (Serverless)	AWS Lambda for API functions, S3 for storing document PDFs and photos, Cognito for secure identity management.

7.2 Data Security and Availability

- **Encryption:** All data at rest is encrypted using **AES-256**. Data in transit uses **TLS 1.3**.
- **Offline First Architecture:** The mobile apps will utilize a local database (e.g., **WatermelonDB** or **SQLite**). All changes (squawks, time logs) are written locally first. A "Sync Manager" runs in the background, pushing changes to the cloud when connectivity is restored and handling conflict resolution (e.g., if two users edit the same record offline).³⁴

8. Detailed Database Schema Design

To support the requirements, the database must map the relationships between aircraft, users, tasks, and external CAMP data.

8.1 Entity Relationship Diagram (ERD) Overview

- **Aircraft_Master:**
 - PK_TailNumber, Serial_No, Model_Type, Total_Hours, Total_Cycles, CAMP_System_ID, Current_Status (Green/Amber/Red).
- **Digital_MEL_Library:**
 - PK_MEL_ID, Aircraft_Model, ATA_Chapter, Item_Title, Repair_Category (A/B/C/D), Procedures_Text_O, Procedures_Text_M.
- **Work_Order_Header:**
 - PK_WO_ID, FK_Aircraft_ID, Status (Open/Closed), Priority (Routine/AOG), Assigned_Team_ID.
- **Task_Detail:**
 - PK_Task_ID, FK_WO_ID, Source (CAMP/Manual), CAMP_Task_Reference, Description, Estimated_Duration (AI_Generated), Status, Handover_Note.
- **Flight_Log (The Squawk Sheet):**
 - PK_Log_ID, FK_Aircraft_ID, Pilot_User_ID, Flight_Phase, Squawk_Description, MEL_Ref_ID, Signature_Hash.
- **Time_Tracking_Log:**
 - PK_Time_ID, FK_Task_ID, FK_User_ID, Start_Time, End_Time, Duration_Minutes, Type (Work/Pause).

9. Conclusion and Implementation Roadmap

This report outlines a comprehensive path to building a market-leading aviation maintenance platform. By satisfying the user's requirements for **TrustFlight-like** usability, **CAMP** data integration, and **predictive intelligence**, the system delivers substantial value:

1. **Regulatory Safety:** Automated compliance through digital MELs and AC 120-78B e-signatures.
2. **Operational Efficiency:** Reducing AOG time via predictive resource allocation and streamlined shift handovers.
3. **Data Sovereignty:** Owning the complete maintenance history (manual + CAMP data) for better asset value retention.

Recommendations for Next Steps:

- **Phase 1 (MVP):** Build the Pilot ETL and Manager Dashboard. Secure FAA "Letter of Intent" for electronic recordkeeping.
- **Phase 2 (Integration):** Develop the CAMP API middleware and Technician Mobile App with shift handovers.
- **Phase 3 (Intelligence):** Deploy the Machine Learning models once 3-6 months of training data has been accumulated.

This architecture ensures the solution is not just a "logbook app," but a sophisticated enterprise asset management platform tailored for the specific, rigorous demands of corporate aviation.

Citations Used

- **Regulatory Compliance:** ¹
- **Operational Workflow & TrustFlight Features:** ¹¹
- **CAMP Systems Integration:** ¹⁸
- **MEL & Deferral Process:** ¹³
- **Predictive Analytics & Algorithms:** ²⁵
- **Shift Handover & Human Factors:** ³
- **Technical Architecture:** ³¹

Works cited

1. AC 120-78 - Acceptance and Use of Electronic Signatures, Electronic Recordkeeping Systems, and Electronic Manuals (Cancelled) - FAA, accessed February 14, 2026, https://www.faa.gov/regulations_policies/advisory_circulars/index.cfm/go/document.information/documentid/23224
2. AC 120-78B - Advisory Circular, accessed February 14, 2026, https://www.faa.gov/documentLibrary/media/Advisory_Circular/AC_120-78B_FAA_Web.pdf
3. Digital Shift Handover Logbook for Airports: Improving Maintenance & Operations

- Coordination - Oxmaint, accessed February 14, 2026,
<https://oxmaint.com/industries/aviation-management/airport-digital-shift-handover-logbook-operations>
4. AC 120-78B - Electronic Signatures, Electronic Recordkeeping, and Electronic Manuals, accessed February 14, 2026,
https://www.faa.gov/regulations_policies/advisory_circulars/index.cfm/go/document.information/documentID/1043396
 5. Advisory Circular AC 120-78A - FAA, accessed February 14, 2026,
https://www.faa.gov/documentlibrary/media/advisory_circular/ac_120-78a.pdf
 6. 1. What is the purpose of this advisory circular (AC)? a. This AC is not mandatory and does not constitute a regulation. Th - FAA, accessed February 14, 2026,
https://www.faa.gov/documentlibrary/media/advisory_circular/ac_120-78_final.pdf
 7. 14 CFR Part 145 Subpart E -- Operating Rules - eCFR, accessed February 14, 2026,
<https://www.ecfr.gov/current/title-14/chapter-I/subchapter-H/part-145/subpart-E>
 8. Requirements for digital pilot logbooks - capzlog.aero, accessed February 14, 2026,
<https://capzlog.aero/academy/regulatory-frameworks/digital-logbook-requirements>
 9. User Guide for CAME (CAMO-AOC) UG.CAO.00162-002 - EASA, accessed February 14, 2026, <https://www.easa.europa.eu/en/downloads/136218/en>
 10. Regulations | EASA - European Union, accessed February 14, 2026,
<https://www.easa.europa.eu/en/the-agency/faqs/regulations>
 11. Tech Log | TrustFlight, accessed February 14, 2026,
<https://www.trustflight.com/products/electronic-tech-log/>
 12. Electronic Tech Logs & how they can improve your operation - TrustFlight, accessed February 14, 2026,
<https://www.trustflight.com/resource/electronic-tech-logs-how-they-can-improve-your-operation/>
 13. AC 120-MEL - Federal Aviation Administration, accessed February 14, 2026,
https://www.faa.gov/sites/faa.gov/files/aircraft/draft_docs/afs_ac/AC_120-MEL_Coord_Copy.pdf
 14. eTechlog - Flightman, accessed February 14, 2026,
<https://www.flightman.com/etechlog/>
 15. MEL Manager - TrustFlight, accessed February 14, 2026,
<https://www.trustflight.com/products/mel-manager/>
 16. AC 91-67A - Advisory Circular - FAA, accessed February 14, 2026,
https://www.faa.gov/documentLibrary/media/Advisory_Circular/AC_91-67A.pdf
 17. MEL Management in Aviation: From Compliance to Supply Chain Strategy - ePlaneAI, accessed February 14, 2026,
<https://www.eplaneai.com/blog/mel-management>
 18. CAMP MTX Maintenance Management, accessed February 14, 2026,
<https://www.campsystems.com/maintenance>
 19. CAMP MTX Web API, accessed February 14, 2026,
<https://icampapp.campsystems.com/webapi/>

20. CAMP Connect - corridor – erp, accessed February 14, 2026,
<https://www.corridor.aero/wp-content/uploads/2024/08/CAMP-Connect.pdf>
21. CAMP work orders/inventory : r/aviationmaintenance - Reddit, accessed February 14, 2026,
https://www.reddit.com/r/aviationmaintenance/comments/1fld0o9/camp_work_ordersinventory/
22. eWorkOrder - CAMP Systems, accessed February 14, 2026,
<https://www.campsystems.com/eworkorder>
23. Effective Shift Handover - Honeywell Process Automation, accessed February 14, 2026,
https://process.honeywell.com/content/dam/process/en/documents/document-lists/doc_asm-consortium/white-papers/July%2031%202011%20-%20Effective%20Shift%20Handover.pdf
24. Free Download Line Maintenance Shift Handover Checklist - Meegle, accessed February 14, 2026,
https://www.meegle.com/en_us/advanced-templates/aircraft_maintenance/line_maintenance_shift_handover_checklist
25. Prediction of Non-Routine Tasks Workload for Aircraft Maintenance with Supervised Learning - TU Delft Repository, accessed February 14, 2026,
https://repository.tudelft.nl/file/File_f6edf3fa-767a-402d-92aa-a83e7353b61e
26. Aircraft Predictive Maintenance: Insight Accelerator | Boeing Global Services, accessed February 14, 2026,
<https://services.boeing.com/maintenance-engineering/maintenance-optimization/insight-accelerator>
27. Bayesian Component Reliability Estimation: an F-35 Case Study - Institute for Defense Analyses, accessed February 14, 2026,
<https://www.ida.org/-/media/feature/publications/b/ba/bayesian-component-reality-estimation-an-f-35-case-study/d-10561.ashx>
28. Aircraft Predictive Maintenance: Benefits of Real-Time Data - Striim, accessed February 14, 2026, <https://www.striim.com/blog/predictive-aircraft-maintenance/>
29. How to Use Predictive Maintenance to Decrease Costs in Aviation - ePlaneAI, accessed February 14, 2026,
<https://www.eplaneai.com/blog/how-to-use-predictive-maintenance-to-decrease-costs-in-aviation>
30. What statistical approaches can optimize maintenance schedules based on historical aircraft failure data? | ResearchGate, accessed February 14, 2026,
https://www.researchgate.net/post/What_statistical_approaches_can_optimize_maintenance_schedules_based_on_historical_aircraft_failure_data
31. SSA UI kit: The ultimate React JS dashboard solution, accessed February 14, 2026,
<https://www.ssa.group/ssa-ui-kit-react-js-dashboard/>
32. 16 Best React Dashboards in 2026 | Untitled UI, accessed February 14, 2026,
<https://www.untitledui.com/blog/react-dashboards>
33. WebSocket Protocol Usage in a Real-time Web Application - Visual Craft, accessed February 14, 2026,
<https://www.visual-craft.com/blog/using-websocket-protocol-for-real-time-applications>

[cations/](#)

34. Unlocking the Value of TrustFlight Tech Log, accessed February 14, 2026,
<https://www.trustflight.com/resource/unlocking-the-value-of-trustflight-tech-log/>
35. MEL vs. MMEL: What Aircraft Operators Need to Know - Pilot John, accessed February 14, 2026,
<https://pilotjohn.com/blog/mel-vs-mmml-what-aircraft-operators-need-to-know>
36. Data Analytics in Aircraft Maintenance: From Insights to Action - EXSYN, accessed February 14, 2026,
<https://www.exsyn.com/blog/data-analytics-in-aircraft-maintenance-from-insights-to-action>
37. Software Engineer @ Camp Agency - Teal, accessed February 14, 2026,
https://www.tealhq.com/job/software-engineer_7ea1a64f78e2d2330c5697c34e36ac46ba83f